PeerAhammad M Bagawan

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Profile

Computer Science undergraduate and co-author of an accepted NeurIPS 2026 poster and an arXiv preprint, working on sequential detection, state-space models, and replay in reinforcement learning. Experience in model implementation, experimental evaluation, and ML-powered applications. Seeking a machine learning research or engineering internship.

Education

B.Tech. in Computer Science and Engineering 2023–Expected 2027
Visvesvaraya Technological University (VTU), Belagavi, Karnataka CGPA: 7.2/10 (through Semester 6)

Publications and Manuscripts

- **When Does Sequential Detection Collapse to a Scalar? A Necessary and Sufficient Characterisation.** Prakul Sunil Hiremath, **PeerAhammad M Bagawan**, Sahil Bhekane. *Accepted, NeurIPS 2026 (Poster).* [OpenReview] Contribution: implementation and experiments.
- **PARD-SSM: Probabilistic Cyber-Attack Regime Detection via Variational Switching State-Space Models.** Prakul Sunil Hiremath, Sahil Bhekane, **PeerAhammad M Bagawan**. *arXiv preprint.* [arXiv] [Code] Contribution: model implementation and experiments (CICIDS2017, UNSW-NB15).
- **Early Detection is an Information Problem.** Prakul Sunil Hiremath, Mohammed Kaif, **PeerAhammad M Bagawan**. *Manuscript in preparation.*
- **The Measure Gap: Information-Theoretic Limits of Deterministic Belief Encoders in POMDPs.** Prakul Sunil Hiremath, **PeerAhammad M Bagawan**, Sahil Bhekane. *Manuscript in preparation.*

Research Experience

Research Intern — Capability Dynamics, VTU Sep 2026–Present

- Reviewing literature on emergent and hidden capabilities, elicitation, and evaluation during model learning.
- Identifying research gaps in distinguishing capability acquisition from observable expression under different evaluation conditions; developing a controlled study plan.

Selected Projects

RSD/RAPO — Replay-Suppression Diagnostics and Policy Optimization [GitHub] | [Related preprint]

- Evaluated replay-suppression methods across graph diffusion, grid-world navigation, MovieLens-25M, and Reddit conversation data to study harmful outcomes when previously encountered behaviors recur.
- In repository experiments, reduced the replay-to-exposure harm ratio (AUC-R) by 31.98% on MovieLens-25M ($p = 0.0163$) and the rate of Reddit outcomes scoring ≤ -5 by 77.26% ($p = 2.01 \times 10^{-8}$), compared with stationary baselines.

ThreatFlow — Streaming Cybersecurity Detection [GitHub] | [Live Demo]

- Built a streaming dashboard over chronological CICIDS2017 replay: Random Forest scoring, switching state-space context, incident-level alert aggregation, and a FastAPI/WebSocket backend.
- In the tested configuration, alerts covered 11/11 attack incidents in 3 alert episodes with 0 false episodes.

ECG-TransNet — ECG Arrhythmia Classification and Web Platform [Model] | [Platform]

- Developed a 1D CNN–Transformer for five-class AAMI ECG beat classification on the MIT-BIH Arrhythmia Database, incorporating signal preprocessing, class-imbalance handling, and Grad-CAM/SHAP explainability; measured inference latency below 12 ms on an NVIDIA RTX 3050 GPU.
- Integrated the model into a web platform for interactive ECG-based arrhythmia predictions.

Technical Skills

Languages: Python, JavaScript
ML: PyTorch, JAX, NumPy, Pandas, scikit-learn
Web & Tools: FastAPI, WebSockets, React, Git